Exercise 1.3

1. $\{1,2,3,4,...\}$, $\{1,2,3,4,...\}$, $\{2,4,6,8,...\}$, yes. 2. yes

3.(i) 12 (ii) $4a^2-10a+6$ (iii) 0 (iv) $x^2-7x+12$

4.(i) (a) 9 (b) 6 (c) 6 (d) 0

(ii) 9.5 (iii) (a) $\{x \mid 0 \le x \le 10, x \in R\}$ (b) $\{y \mid 0 \le y \le 9, y \in R\}$

(iv) 5 5. 2 6.(i) -2 (ii) $\frac{3}{2}$ (iii) 3 (iv) $\frac{1}{2}$

7. $4x^3-96x^2+576x$ 8. 1 9. $500t$

10.(i) Yes (ii) 0.9, 24.5 (iii) 60.5 inches (iv) 32 cms

Exercise 1.4

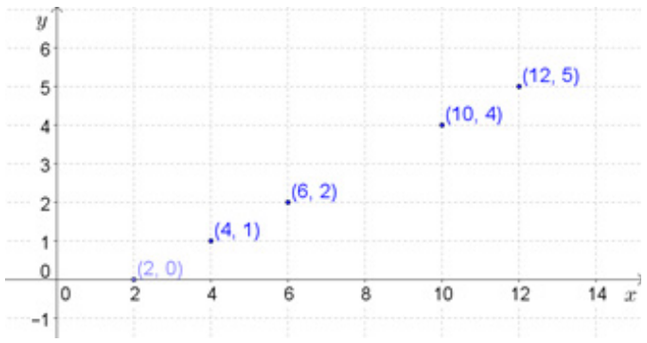
1.(i) Not a function (ii) function (iii) Not a function (iv) function

2.(i) $\{(2,0),(4,1),(6,2),(10,4),(12,5)\}$

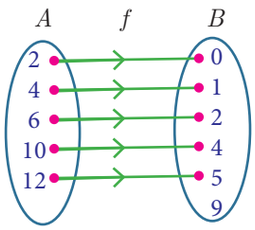
(ii)

x	2	4	6	10	12
$f(x)$	0	1	2	4	5

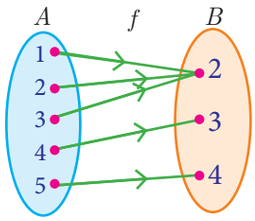
(iv)



(iii)



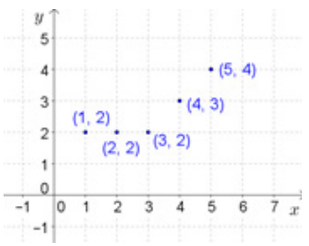
3.(i)



(ii)

x	1	2	3	4	5
$f(x)$	2	2	2	3	4

(iii)



6.(i) $\{1,8,27,64\}$ (ii) one-one and into function

7.(i) Bijective function (ii) Not bijective function 8. a = –1 or 1, b = 1

9.(i) 5 (ii) 2 (iii) -2.5 (iv) 1

10.(i) 2 (ii) 10 (iii) 178 (iv) $\frac{-9}{17}$

11. Yes 12.(i) $32°F$ (ii) $82.4°F$ (iii) $14°F$

(iv) $100°C$ (v) $-40°$

Exercise 1.5

1.(i) x^2-6, $(x-6)^2$; not equal (ii) $\frac{2}{2x^2-1}$, $\frac{8}{x^2}-1$; not equal

(iii) $\frac{3-x}{3}$, $\frac{9-x}{3}$; not equal (iv) $x-1$, $x-1$; equal (v) $4x^2+8x+3$, $4x^2$; notequal

2.(i) -5 (ii) $\frac{-5}{3}$ 4. $a = \pm 2$

5. $\{y \mid y = 2x^2+1, x \in \mathbb{N}\}$; $\{y \mid y = (2x+1)^2, x \in \mathbb{N}\}$ 6.(i) x^4-2x^2

(ii) $\left[x^4-2x^2\right]^2-1$ 7. f is one-one, g is not one-one, $f \circ g$ is not one-one 9. $-4x-1$

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